THE UNIVERSITY OF NEW MEXICO

CS 491/591 INTRODUCTION TO HCI · UBIQUITOUS AND TANGIBLE COMPUTING

Say What You Mean

AI ethics, agentic AI, and what CARC is building, as the way we ask questions of data changes

PRIMER → VIBE AND VOICE → ETHICS → CARC

Tyson L. Swetnam, PhD

Center for Advanced Research Computing · The University of New Mexico

Friday, Nov. 13, 2026

Guest lecture

TODAY

AI ethics, and what CARC is building

01

Primer

Metadata, knowledge graphs, ML, RL, agents, and the devices we ask with

02

Vibe and voice

Coding by conversation and prompting by voice, and the evidence

03

Ethics

Principles, concerns, and who decides

04

CARC

Docs written for AI agents, and resources you can use

One thread: **how we ask questions of data is changing**, and how we say what we mean, by voice or keyboard, steers it.

THIS WEEK: UBIQUITOUS AND TANGIBLE COMPUTING

When the computer disappears, who checks its work?

MARK WEISER · 1991

"The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it."[1]

HIROSHI ISHII & BRYGG ULLMER · 1997

Tangible Bits: "grasp & manipulate" bits at the center of attention, and sense "background bits at the periphery of human perception."[2]

A voice prompt and an agent are interfaces that disappear: no screen, no menu, sometimes no keyboard. Today is about what that does to **checking the result**.

THE THREAD THROUGH THIS TALK

How we ask questions of data is changing

Query
Write SQL or click filters. You must already know where the data is and what it is called.

→

Search
Type keywords. An engine ranks documents; you still read them and do the analysis.

→

Ask
Type or say a question. A language model answers from what it read in training.

→

Delegate
State a goal. An agent finds the data through its metadata, follows a knowledge graph, runs trained models, checks the result, and reports.

The deeper the question, the more the answer rests on **metadata, knowledge graphs, and models trained by machine learning and reinforcement learning**, and on how well you say what you mean, by voice or keyboard.

01

Primer

The words for how we now ask questions of data, and the devices we ask with.

NORMAN'S GULFS, AND A THIRD FOR LANGUAGE MODELS

AI shrinks one gulf and widens another

ENVISIONING · 2024

Before you ask

Can it do this? How should I say it? How will I judge the answer?[3]

EXECUTION · 1985

Goal → system

Once a menu, a syntax, a manual[4]. Now one sentence, typed or spoken. **The syntax is gone; the articulation is not.**

EVALUATION · 1985

System → goal

Forty changed files, tests you did not write, a confident summary. **Wider than ever.**

Everything in this talk is a tool for crossing the gulf of evaluation.

Six words for asking deeper questions of data

Metadata

"Data about data": what, who, when, where, in what units[5]. FAIR asks that it be usable **by machines**, not only people[6].

Ontology

A formal, machine-readable statement of what terms mean[7]: Carabidae, the ground beetles, is a family within Coleoptera, the beetles[8].

Knowledge graph

Things as nodes, relations as edges: specimen → *collected at* → site[7]. GraphRAG lets a language model answer questions across one[9].

Machine learning

A system learns if its performance at a task improves with experience[10]; supervised models learn from labeled examples to classify, detect or segment.

Reinforcement learning

Learning what to do by trial and error to maximize a reward[11]. It tunes chatbots to follow instructions[12] and trains reasoning and coding agents[13,14].

Agentic AI · harness

A language model using tools in a loop, choosing its next step from what comes back[15]. The harness is the app you talk to: Claude Code, Codex, Antigravity.

One question, every layer

LAYER	WHAT IT DOES	IN THE NEON BEETLE PROJECT
You · prompt	Say or type the question and what counts as done	“Which ground beetle is most common at the Jornada site? Show your checks.”
Agent · LLM trained with RL	Plans steps, calls tools, writes SQL, checks results	Claude, reading the lake read-only over WebDAV
Knowledge · metadata and ontologies	Finds the right data and knows what its columns mean	Columns tagged with ontology terms: NCBITaxon for taxa, UO for units, COLAO for the elytron[16]
Models · conventional ML	Turns photos into measurements	Grounding DINO finds each beetle, SAM 2.1 outlines it, a Mask2Former model labels head, pronotum and elytra[16]
Data · lakehouse	Stores tables with history	DuckLake tables on the CyVerse Data Store, queried with SQL

Ontology-linked metadata is the building block of a knowledge graph; MESA does not yet ship a knowledge-graph product.

Every device is becoming a way to prompt

DEVICE	IT USED TO	NOW
Keyboard	Type commands, code and text	Write prompts and approve steps in a harness: Claude Code in a terminal, IDE, desktop app or the web[17]; Antigravity[18]
Mouse and display	Point, click, read the screen	An agent clicks and types for you while you review: Claude Desktop[19] and ChatGPT[20] computer use; Claude in Chrome[21], Gemini in Chrome[22], Comet[23]
Microphone	Phone calls, occasional dictation	Dictate prompts into Claude Code; full spoken conversations in Claude's voice mode[24] and ChatGPT Voice[25]
Camera	Take pictures	Point the phone at something and talk about it with Gemini Live[26]
Phone and tablet	Run apps you operate	A remote for agents that keep working: Claude Code Remote Control[27], Codex on iOS[28]; always-on agents that message you: Claude Duet[29], OpenAI's dots[30]

The screen is becoming the place you **review and approve**, not the place you operate.

02

Vibe and voice

Coding by conversation, prompting by voice, and knowing when to stop trusting the vibe.

The first vibe coder was talking, not typing

> **"There's a new kind of coding I call 'vibe coding', where you fully give in to the vibes, embrace exponentials, and forget that the code even exists."**

THE SAME POST

"I just talk to Composer with SuperWhisper so I barely even touch the keyboard. ... I 'Accept All' always, I don't read the diffs anymore."

Andrej Karpathy, post on X[31]

NINE MONTHS LATER

Collins Word of the Year 2025: "the use of artificial intelligence prompted by natural language to write computer code."

Collins Dictionary[32]

THE GULF OF EVALUATION, MEASURED

Feeling faster is not being faster

19%

longer to finish real issues with early-2025 AI tools

16 experienced open-source developers, 246 issues. They forecast a 24% speedup, and afterwards still believed in a 20% one[33].

Tools improve fast: by Feb. 2026 METR called its follow-up an unreliable signal, because more developers declined to work without AI[34]. The gap between feeling and measurement is the point.

5.2% and 21.7%

at least, on average: the share of packages in code from commercial and open-source models that did not exist. 205,474 unique invented names in 576,000 samples, each one a name an attacker can register[35].

45%

of cases introduced a security vulnerability, across 80 coding tasks and more than 100 models. A vendor's own report, not peer reviewed[36].

THE THREE TOOLS I USE

Where your voice goes matters

	CLAUDE CODE `/voice`	WISPR FLOW	SUPERWHISPER · MACWHISPER
Where	Inside Claude Code: hold Space to talk, or `/voice tap`	Any app you can type in	Any app, as system-wide dictation
Transcribed	On Anthropic's servers; needs a Claude.ai sign-in	Always in the cloud; audio deleted afterwards unless you keep it	**On your Mac**, with local models such as Whisper[37]; cloud optional
Adds	Nothing to install; 20 languages	Drops filler words, punctuates, tags file names in Cursor and Windsurf	Custom modes with your own instructions; file and meeting transcription
I use it for	Talking to the agent already open	Long briefs, email, review comments	Anything sensitive: unpublished data, student records

Sources: Claude Code[38], Wispr Flow[39,40], superwhisper[41], MacWhisper[42]. Features and privacy terms are the vendors' own statements, as of Nov. 2026.

A SPOKEN PROMPT IS A BRIEF

Say the context; type the identifiers

TYPED IN A HURRY · 7 WORDS

```
add time travel to the beetle lake
```

SPOKEN · 79 WORDS · ILLUSTRATION, NOT DATA

```
In the beetle seg repo, add a time travel query to the duck lake metadata
history, not the measurements. Use the existing client; no new
dependencies. Take a collection path and a date, and return the a views as
they were on that date. Read only: nothing writes to eye rods. Add a test
that uses last week's snapshot, run the linter, and only then tell me it's
done. If the snapshot has expired, say so instead of guessing.
```

Check the nouns

Dictation fails on names: DuckLake (data format), AVUs (metadata tags), iRODS (data store). Misheard words are underlined.

Read before Enter

A misheard word becomes an instruction the agent will follow.

Pin down what must be exact

Type, spell out, or add to a custom vocabulary: paths, identifiers, numbers, code.

03

Ethics

Principles, concerns, and who decides.

Many AI principles are interface requirements

FLORIDI & COWLS · 2019

Five principles for AI in society

Beneficence · Non-maleficence · Autonomy · Justice · **Explicability**[43]

Explicability, being able to see how a decision was made and who is responsible, is what makes the other four checkable.

WHITE HOUSE OSTP · OCT. 2022

Blueprint for an AI Bill of Rights

Safe and Effective Systems · Algorithmic Discrimination Protections · Data Privacy · **Notice and Explanation** · **Human Alternatives, Consideration, and Fallback**[44]

Non-binding. The two in bold are design requirements: tell people an automated system is in use, and let them reach a person.

HCI is where principles become interfaces: a disclosure, an explanation, a way to reach a human.

Serious questions deserve serious answers

Power and water

Data-centre electricity is projected to more than double by 2030, to about 945 TWh, about what Japan uses today[45]. Cheaper AI may mean more AI, as Jevons saw with coal[46].

Consent and credit

Courts have held training on books to be fair use, but pirated copies went to trial: Bartz v. Anthropic settled for $1.5 billion, about $3,000 per work[47].

Bias

A widely used US care algorithm steered less care to Black patients because it used past health spending as a proxy for need[48].

Accountability

In the July 2026 OpenAI–Hugging Face incident, roughly 7% of agent transcripts carried spoofed tool calls; an independent investigation established what happened[49].

Each could be its own lecture. GPT 101's ethics module covers them in depth[50].

AI ETHICS · THE 2026 DEBATE

Three 2026 essays: build capacity before the disruption

RUTGER BREGMAN · JUNE 2026

Don't abandon the field

"AI denial" is the left's climate denial: "abandoning the field is not the same as stopping the technology." A lab's restraint is no substitute for public institutions: "conscience is not a policy"[51].

BILL GATES · AUG. 26, 2026

We are not preparing

The transition "will be one of the most turbulent times in human history … Right now, we are not preparing for it." He proposes "Human Reserved" jobs and taxing robots like hiring[52].

DARIO AMODEI · SEPT. 2026

Pace, don't pause

"We must slow the pace at which we improve the capabilities of AI models," with outside evaluators embedded at the lab, free to publish[53].

All three reject "shut it down", and all three ask for capacity to be built **before** the disruption arrives.

GUARDRAILS ARE INTERFACE DESIGN

Make hallucination checkable

THE OBJECTION	WHAT IS MISSING	THE CONTROL
“It invents citations”	No retrieval step at all	The agent must call a real index
“It makes up metadata”	Open vocabulary, nothing to check	Terms must resolve in an ontology
“It varies run to run”	Sampling with no output contract	A JSON schema fixes the shape; the other controls check the content
“I cannot audit it”	No record of inputs or steps	Provenance in DuckLake; replay the tool calls

Each control was demonstrated with code and data in the Nov. 4 colloquium’s evidence repository[54].

WHO WRITES THE RULES FOR AI?

“The future of #ArtificialIntelligence is not pre-determined. As a technology developed by human beings, its future depends on the choices we make.”

Pope Leo XIV, Oct. 30, 2026[55]

“Red states and blue states are writing AI Bills of Rights.” Alondra Nelson, May 2026[56]. Sitting it out is also a choice.

04

CARC

Documentation for people and for AI agents, and resources you can use today.

AGENTIC, INTERACTIVE DOCUMENTATION · LLMS.TXT[57]

CARC's docs have a front door for agents

```
# CARC Documentation

> User and researcher documentation for the UNM Center for
> Advanced Research Computing (CARC): high-performance computing,
> storage, software, tutorials, and training. …

## Getting started
- [Getting started at CARC](…/docs/getting-started/overview/):
  Create a CARC account, join a project in ColdFront, and find
  support. Markdown twin: …/overview/index.md
  Raw source: raw.githubusercontent.com/…/overview.md
…
## Meta
- [Full corpus in one file](…/docs/llms-full.txt): every page's
  Markdown with frontmatter, links made absolute; about 696 KB,
  roughly 178,000 tokens.
```

Why

HTML wraps information "in navigation, ads, and JavaScript," and context windows are still too small for most whole sites[58].

Three addresses

All ~80 pages: the rendered HTML, a Markdown twin (+`index.md`), and the raw source on GitHub, for sandboxes that block the site[59].

Same convention

MESA[60], this course site, DUST 2026[61] and GPT 101 publish the same way.

OPEN KNOWLEDGE FORMAT V0.2[62]

Frontmatter is an interface, too

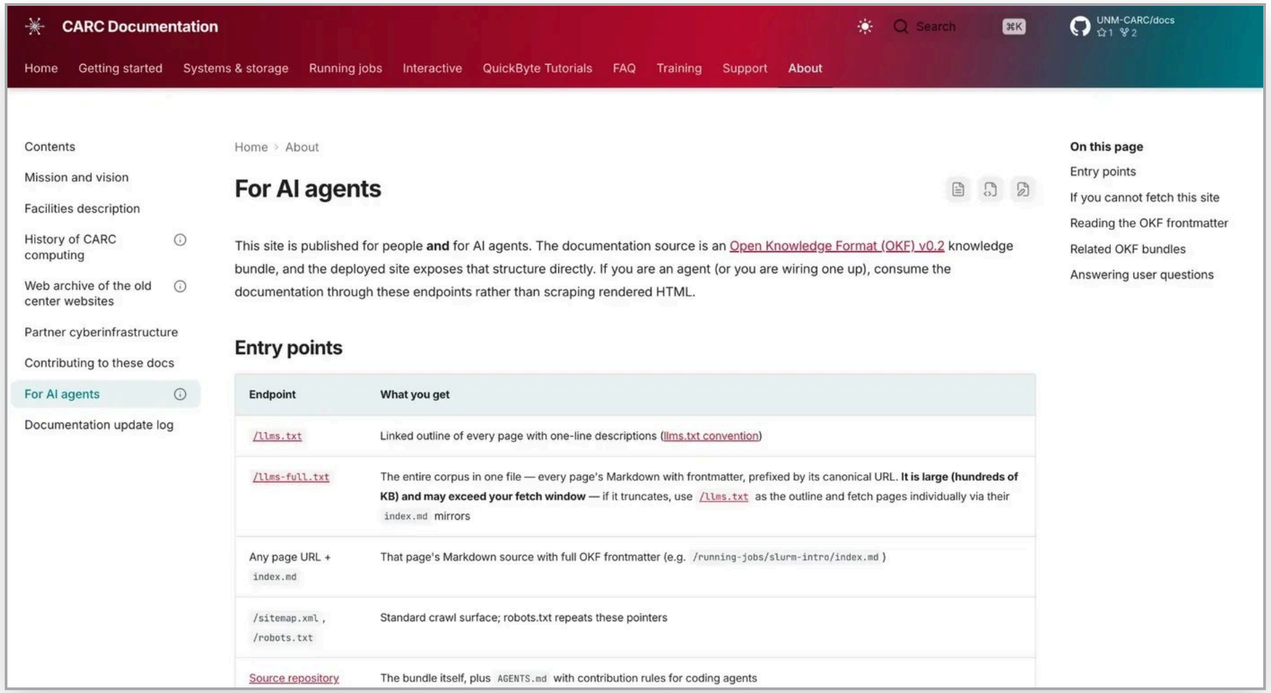
```yaml
---
title: "For AI agents"
description: "How agents and harnesses …"
type: Reference
generated:
  by: "claude/fable-5-1"
  at: "2026-09-12T00:00:00Z"
sources:
  - id: okf-spec …
  - id: llmstxt …
status: stable
# no verified: key, so this page is unverified
---
```

THE READER ASKS	THE PAGE ANSWERS WITH
What is this?	`type`, `title`, `description`
Where did it come from?	`generated`, `sources`
How much should I trust it?	`verified` → unverified, machine-confirmed or human-reviewed
Is it still true?	`status`, `stale_after`

CARC's agent guide says an AI drafted it and no person has verified it yet[63]. Disclosure is an ethics feature.

TRY IT · WITH A HUMAN IN REACH

Paste our URL into a chat



`/llms.txt`: an outline of all ~80 pages[59].

`/llms-full.txt`: the whole corpus, about 696 KB, roughly 178,000 tokens.

“When the corpus does not answer a question, direct users to the humans … do not guess cluster-specific facts”[63].

> “Read carc.unm.edu/docs/llms.txt, then tell me how to get access to the H100 GPUs on Easley. Cite the pages.”

Works where the chat tool is allowed to fetch URLs; for sandboxes that block it, llms.txt points to raw GitHub copies.

THE MESA PATTERN

Docs teach the agent how; tools fetch the data

1 · Speak
A researcher dictates the question and its constraints.

→

2 · Read
The agent reads `llms.txt` and learns which servers to install.

→

3 · Connect
MCP's `tools/list` returns the live, authoritative tool catalogue.

→

4 · Query
Queries run where the data lives: read-only anonymously, with your rights once you sign in.

→

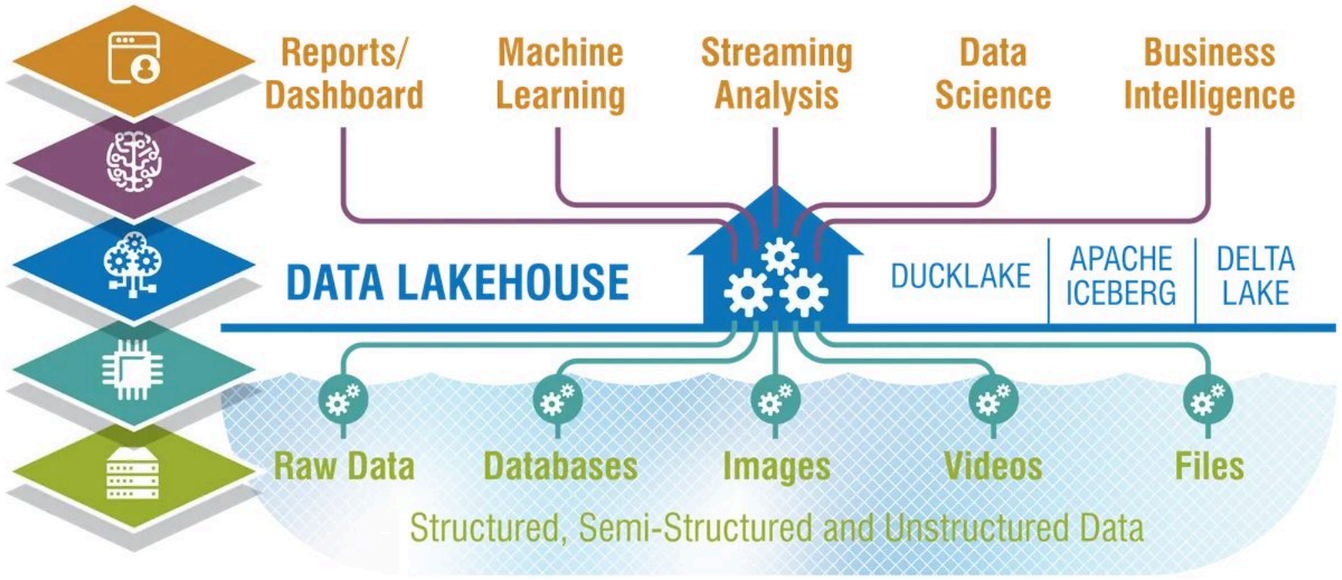
5 · Record
Each metadata write through mesa-mcp becomes a change record with provenance.

> **"If what you actually want is CyVerse data … do not scrape these pages: connect the servers and call their tools."**

MESA documentation, For AI agents[64]

DATA LAKEHOUSE

A lakehouse that remembers



Lakehouse: cheap, directly accessible storage plus database features: safe concurrent updates (ACID transactions), versioning, auditing[65].

DuckLake: metadata in an ordinary SQL database, data in open Parquet files[66]; 1.0 in April 2026[67].

In MESA: every metadata-tag write through mesa-mcp (an iRODS attribute-value-unit, or AVU) is an append-only change record in DuckLake snapshots, so an agent can ask what a collection's metadata was as of last Tuesday[68,69].

Ships today: AVU metadata history in DuckLake. Planned under the award, 2026–2029: the full lakehouse and a federated iRODS data mesh[70].
Figure: MESA proposal.

RESOURCES YOU CAN USE

What CARC offers students and faculty

GPUs and clusters

Easley: 8 NVIDIA H100 and 36 L40S GPUs. Hopper: 26 A100 and 2 V100[71]. The H100 and L40S partitions need an allocation your PI requests[72].

Free access

Free to UNM researchers: create an account, and a PI adds you to a project in ColdFront[73]. No regulated data (HIPAA, FERPA, CUI) on CARC systems[72].

People

Tickets at support.alliance.unm.edu, email help@carc.unm.edu, office hours and consultations[74].

Training

Semester workshops, domain workshops and course guest lectures[75]; open lessons on AI ethics in DUST 2026[61] and GPT 101[76].

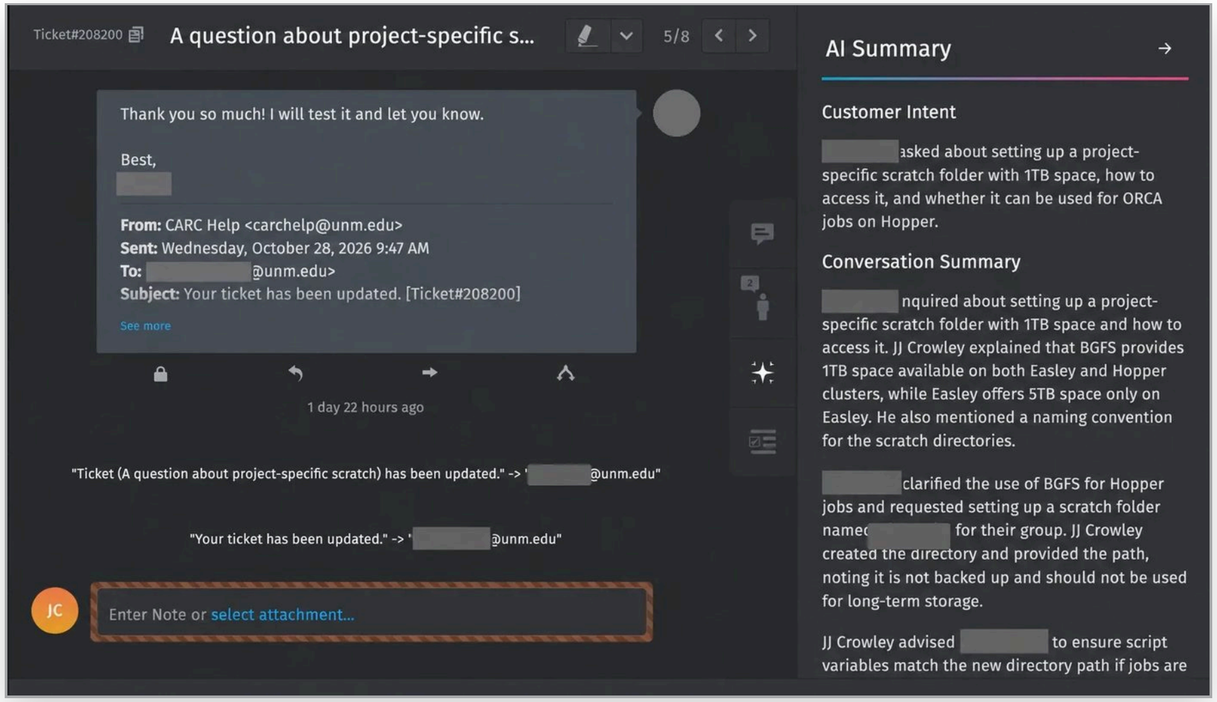
Docs for agents

`carc.unm.edu/docs/llms.txt`, Markdown twins and a full corpus, with a human fallback[63].

Hosted AI: not yet

CARC "does not currently provide hosted AI, large language model (LLM), or agent services as a production service"; it points to the NAIRR Pilot, ACCESS and CyVerse AI-VERDE[77].

Self-hosted AI on our help desk



Tickets, summarized: staff see what the person needs and what has happened so far.

Open weights, our hardware: in our setup, an open-weight model on two NVIDIA DGX Spark systems, each with 128 GB of unified memory[78]; tickets are not sent to an outside AI service.

A person still answers. The summary helps staff; it does not reply for them.

A real ticket in CARC's Zammad help desk, Oct. 28, 2026; requester details redacted. Setup as reported by CARC[54].

Design for the gulf of evaluation

01

Design for two readers

Publish `llms.txt`, Markdown twins and frontmatter next to the HTML.

02

Make checking cheap

Tests, validators, provenance and trust signals, so checking costs less than redoing.

03

Keep a human in reach

Disclose the AI, explain it, and give people a path to a person, as the Blueprint asks.

04

Engage, with care

Opting out is also a choice. Learn the tools with CARC's free resources.

The question is becoming the interface to data. Design the data's description, the prompt and the check together.

THANK YOU, PROFESSOR BUECHLEY AND CS 491/591

Questions?

READ IT THE WAY AN AGENT DOES

idss-mesa.github.io/docs/llms.txt

carc.unm.edu/docs/llms.txt

ACKNOWLEDGMENTS

MESA is supported by the U.S. National Science Foundation under Award OAC-2632685[70].

REUSED SLIDES

From the UNM Computer Science Colloquium, Nov. 4, 2026, with public evidence for every slide[54].

Tyson L. Swetnam, PhD

tswetnam@unm.edu

Center for Advanced Research Computing · The University of New Mexico

SLIDES & REFERENCES

tyson-swetnam.github.io/AI-Automation-and-Agents

Backup

Going deeper: HCI principles for agents, three slides from the Agentic AI Loops lecture, MESA in more depth, and two activities for the class.

THREE UI PARADIGMS

From commands to outcomes

FROM ABOUT 1945

Batch

Hand over the whole job, come back for the printout. No interaction at all.

FROM ABOUT 1964

Command-based

Take turns, one command at a time. Command lines and graphical interfaces both live here. Nielsen files **direct manipulation**[79] here too, though its inventors framed it as a step beyond commands.

NOW

Intent-based outcome specification

Say the result you want, not the steps. It “completely reverses the locus of control.”

Nielsen calls the third the first new UI paradigm in 60 years[80]. **Vibe coding is intent-based interaction aimed at software.**

THE 1997 BET

“Direct manipulation interfaces are seen as more likely candidates to influence advanced user interfaces than adaptive, autonomous, intelligent agents. User control and responsibility are highly desirable.”

Ben Shneiderman, 1997[81]

Agents have arrived in the interface. His second sentence is still the design problem.

PLAIN-LANGUAGE DEFINITIONS

Model, agent, harness, protocol

The model

A language model answers from what it read in training. It cannot see your files or run your code.

The agent

A loop around the model: read the goal, call a tool, read the result, decide the next call.

The harness

The program you actually run, such as Claude Code or Codex. It owns the loop, the files it may touch and the permissions.

The protocol

MCP, open since Nov. 2024[82]. Its docs call it “a USB-C port for AI applications”[83]: build a server once, and any agent that speaks MCP can use it.

A DIAL, NOT A SWITCH

Vibe the prototype; engineer what others depend on

Vibe coding: accept every diff

Agentic engineering: orchestrate, review, test

SIMON WILLISON · MAR. 2025

“If an LLM wrote the code for you, and you then reviewed it, tested it thoroughly and made sure you could explain how it works to someone else that’s not vibe coding, it’s software development.”[84]

KARPATHY · FEB. 2026

His new favorite name, “agentic engineering”: “you are not writing the code directly 99% of the time, you are orchestrating agents who do and acting as oversight.”[85]

SARKAR & DROSOS · PPIG 2025

What the authors call the first empirical study of vibe coding, from 8+ hours of think-aloud video: expertise is not eliminated but moves to “context management, rapid code evaluation,” and knowing when to take over by hand[86].

HOW WE KEEP VIBE-CODED REPOSITORIES HONEST

Constraints and feedback, written for the agent

Contract

AGENTS.md and **CLAUDE.md** say what the agent may edit, which files a script owns, and what it must never write[87].

Constraints

Closed vocabularies: fourteen page types, and ontology terms looked up in the Ontology Lookup Service, with a curator approving any mapping.

Feedback

Every push runs the validators: format conformance, a course lint, a strict build. The agent reads the errors and fixes them.

Evidence

Every number on my last talk's slides has an evidence entry: code to rerun, data to trace, and a status, including "not yet checked"[54].

Norman's design principles still apply. The user is now an agent: it is handed the manual every session, and the validators check whether it followed it.

WHY SPEAK A PROMPT?

Speaking beat thumb-typing about 3× when copying short phrases

2.9×

English speech vs. a phone keyboard, copying given phrases: 153 vs. 52 words per minute

Fewer errors while entering, slightly more left in the final text. A lab transcription task on phones, not composition; nobody has yet measured prompts to coding agents[88].

The articulation barrier

Prompting asks people to be “highly articulate” in written prose[89]. Speaking “probably alleviates the articulation barrier somewhat”[90].

Access, not convenience

For programmers with motor impairments, voice is the way in. They want to speak code naturally, not memorize unnatural command grammars[91].

HARNESSES AND ASSISTANTS

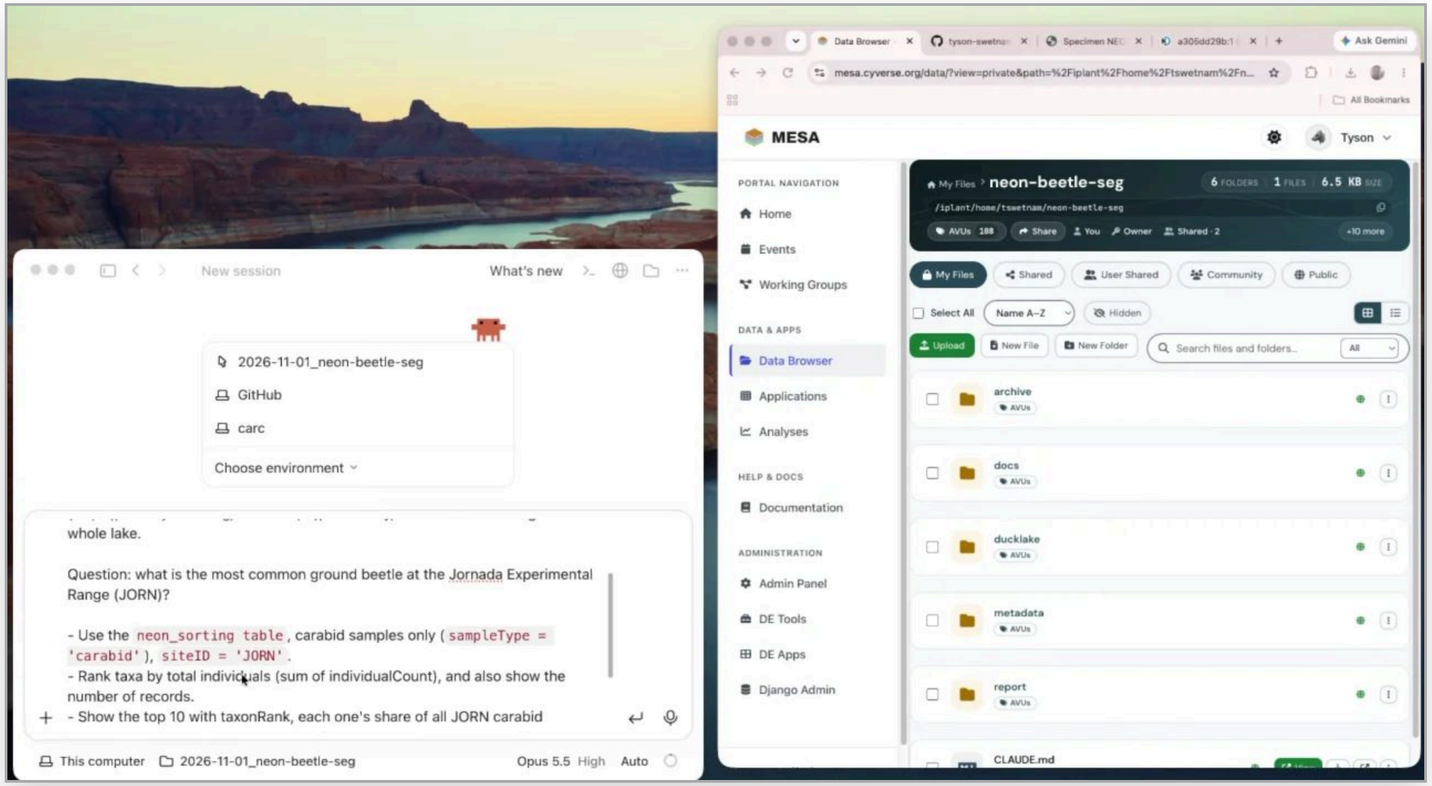
Who makes what, November 2026

VENDOR	CHAT AND VOICE	CODING HARNESS	ACTS IN A BROWSER OR DESKTOP	ALWAYS-ON AGENT
Anthropic	Claude apps, voice mode[24]	Claude Code: CLI, IDE, desktop, web[17]	Claude in Chrome[21]; computer use in Claude Desktop[19]	Claude Duet, research preview[29]
OpenAI	ChatGPT, ChatGPT Voice[25]	Codex: CLI, IDE, cloud, iOS[28]	Computer use in the ChatGPT desktop app[20]	dots[30]
Google	Gemini, Gemini Live[26]	Antigravity: desktop app, CLI, SDK[18]	Gemini in Chrome, auto browse[22]	not covered here
Perplexity	not covered here	not covered here	Comet browser: Mac, Windows, iOS, Android[23]	not covered here

Vendors' own descriptions, checked Nov. 13, 2026. Products change monthly; "not covered here" means not checked for this talk, not absent.

DEMO · RECORDED NOV. 2, 2026 · 2:10 AT 1.5×

From a question to a result



Asked: the most common ground beetle at NEON's Jornada site, from the neon-beetle-seg DuckLake, read-only, without downloading the lake[92].

The agent's answer: ***Tetragonoderus pallidus***, 616 of 1,056 carabid individuals (58.3%), with its data checks shown.

Ran on the speaker's computer against the CyVerse Data Store. An independent recompute from NEON is still pending[54]. Watch the recording.

Old principles for new agents

PRINCIPLE	IN A CODING AGENT
“Make clear what the system can do” (G1)[93]	The harness shows the person the agent’s tools and permissions before it acts
“Make clear how well the system can do what it can do” (G2)[93]	Status, sources and the verified tier in every page’s frontmatter
“Support efficient correction” (G9)[93]	Small diffs, one commit per step, plan before edit, easy revert
Employ dialog to resolve key uncertainties[94]	The agent asks before acting on an ambiguous request, weighing the cost of interrupting
Rapid, incremental, reversible operations[81]	Git history for code; append-only DuckLake snapshots for metadata
Acceleration vs. exploration[95]	Know which mode you are in: you know the next step, or you are asking the agent to show you options

Who decides what happens next

BUILDING BLOCK	IN ORDINARY SOFTWARE	IN AN AI AGENT
Sequence	A programmer wrote the steps, in order	The AI makes up the steps as it goes
Selection	A programmer wrote every if and else, and anyone can read and test them	The AI chooses, by reading descriptions written in plain English
Iteration	A programmer wrote the rule for when to stop	The AI decides when it is done, unless a step limit cuts it off

The three building blocks are still there. Their author is now an AI, deciding one step at a time[96].

FROM THE AGENTIC AI LOOPS LECTURE

Coding harnesses

HARNESS	FROM	WHERE IT RUNS
Claude Code	Anthropic	Terminal, IDE, web · MCP native
Codex	OpenAI	Terminal, IDE, cloud tasks
Antigravity	Google	Agent-first IDE
OpenCode	Open source	Terminal, model-agnostic
Goose	Open source	Desktop and CLI · MCP native

A harness is your lab bench: a shell, files, version control and MCP servers, with a person approving the steps that matter[96].

FROM THE AGENTIC AI LOOPS LECTURE

AI sandboxes: fences for agents

KIND OF FENCE	WHAT IT DOES	EXAMPLES
A sealed-off computer	The agent works inside a separate machine, so mistakes stay inside it	Virtual machines, containers
Built-in limits	The agent runs on your computer but may touch only certain files, commands and websites	Permission settings in Claude Code[97], Codex, Gemini CLI
Both at once	The agent's own limits, inside a sealed-off machine	Kubernetes pods with traffic through a proxy

Six habits: start with few permissions, review before running, use a dedicated environment, limit file access, watch what it does, keep backups[76].

Thawing public research's frozen data lake

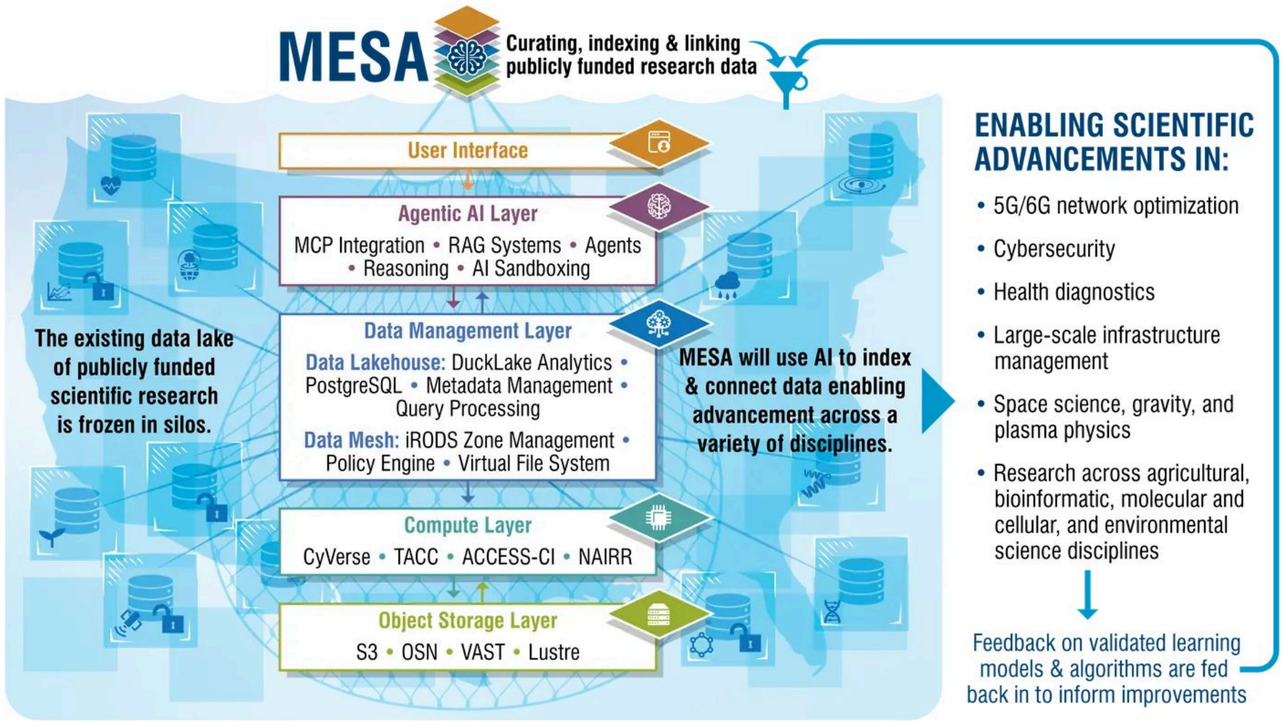


Figure: MESA proposal[98]; the award funds this design through 2029[70].

AGENTIC AI

The agentic flow, as designed



Ask in plain language; the agent finds the data, runs the analysis, and returns a result you can audit.

In the colloquium's evidence this flow ran over real MCP with a scripted plan standing in for the model, and keyword search standing in for retrieval[54]. Figure: MESA proposal.

A DESIGN PATTERN FOR THE DATA LAYER

Give the agent its own account

Read rights only

An agent with its own iRODS account and read access cannot delete, overwrite or change permissions, **whatever its intent**.

Checked before it runs

iRODS policy enforcement points can stop an operation before it executes: who is asking, what rights they hold, which zone and path.

Owners keep control

The owner edits, moves, deletes and shares. Federated zones share data but keep their own catalogs and rules.

A pattern, not what every MESA agent does today: in Demo 1, the agent acts as the signed-in researcher, with that person's rights[54].

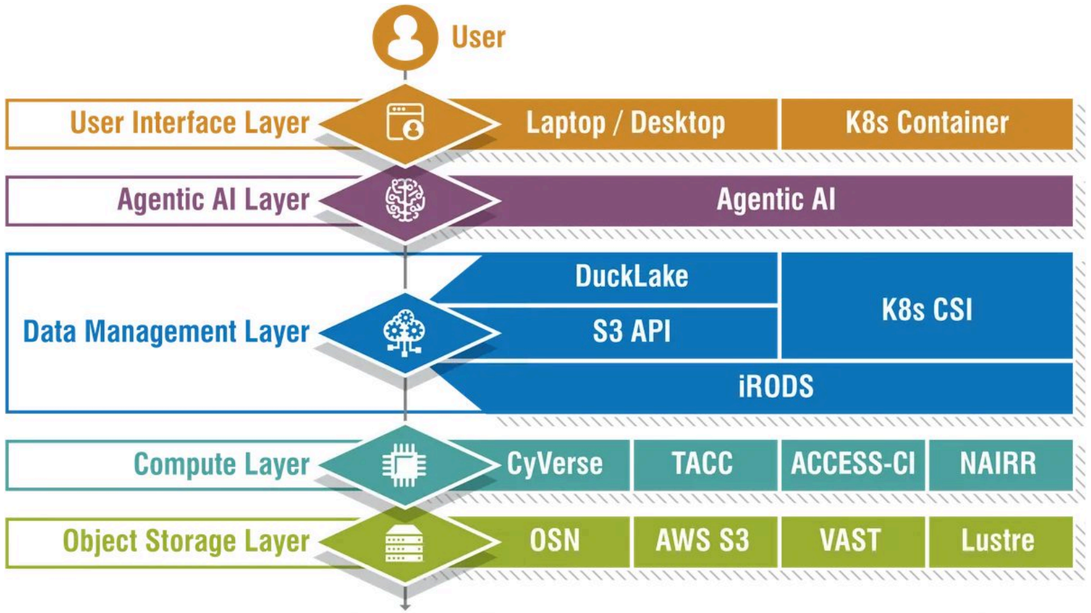
Terms resolve, or go to a curator

FIELD	WHAT FREE TEXT PRODUCES	THE ONTOLOGY TERM
Taxon	ground beetle · carabid · Carabidae	`NCBITaxon:41073` Carabidae
Length	mm · millimeters · millim.	`UO:0000016` millimetre
Area	mm² · sq mm · mm2	`UO:0000082` square millimetre
Specimen	pinned beetle · voucher · sample	`OBI:0100051` specimen, by curator-approved mapping

Exact names and synonyms resolve through the Ontology Lookup Service. “carabid”, “millim.” and “sq mm” do not, so a person approves the mapping.

THE STACK

Five layers, each swappable by design

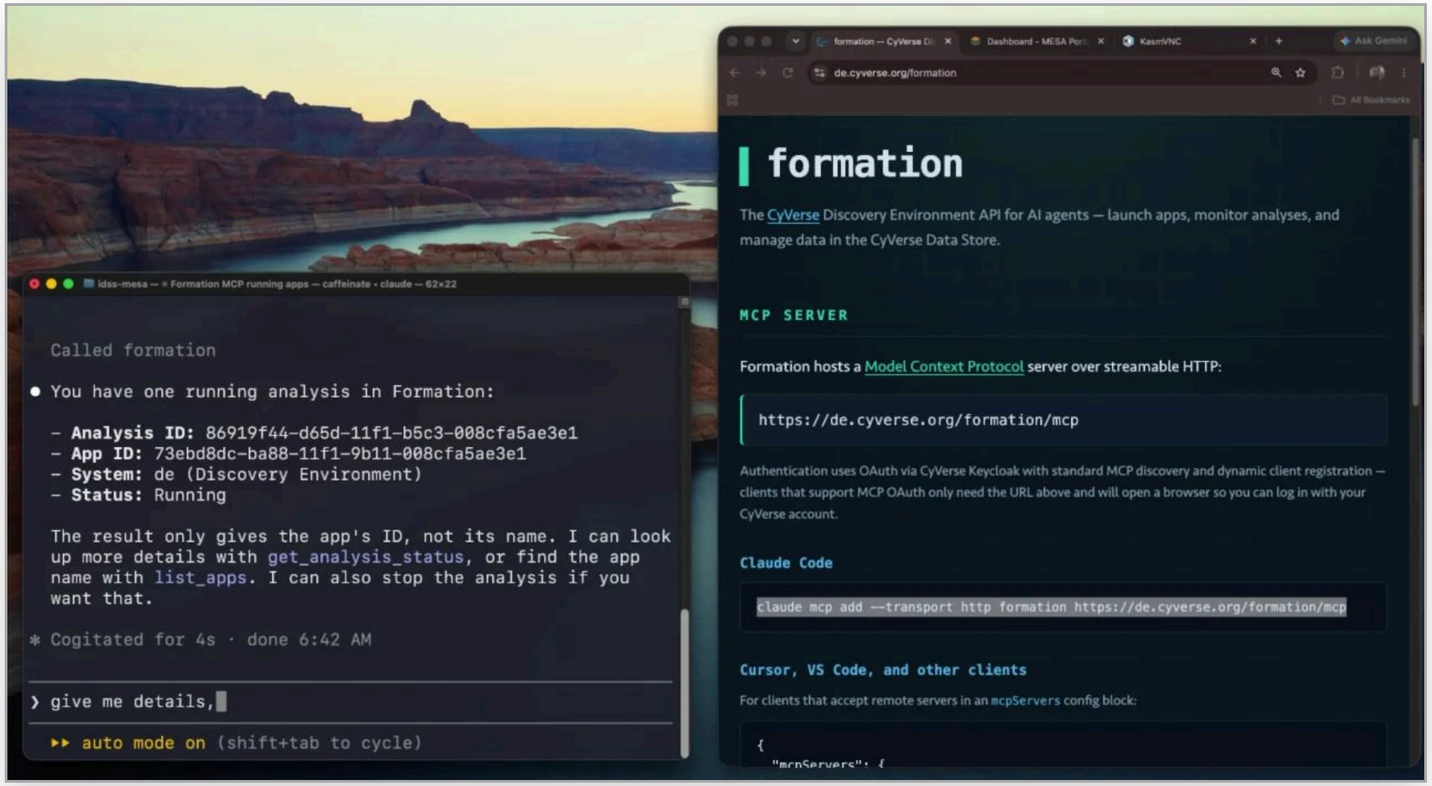


Data Mesh: Data from different domains is managed, coordinated, and presented from different storage technologies.

Open source where it exists (Kubernetes, DuckLake, iRODS, Lustre); some backends, such as VAST and AWS S3, are proprietary. Running today on CyVerse; TACC, Jetstream2 and OSN are planned. Figure: MESA proposal.

DEMO · RECORDED NOV. 2, 2026 · 88 SECONDS

Connect once, then just ask




```
claude mcp add \
  --transport http formation \

https://de.cyverse.org/formation/mcp
```

Sign in with CyVerse in the browser, then ask in plain language: list my running apps; list the folders in my collection.

The agent acts as the signed-in researcher, with that person's rights[54]. Watch the recording.

ACTIVITY · 10 MINUTES · PAIRS

Try it: dictate a brief

1

Pick a task

A small change to something you built this term.

2

Type it

Write the prompt as you normally would. Count the words.

3

Say it

Dictate it with your phone or laptop's built-in dictation, for one minute. Count again.

4

Compare

What did speech add: goal, context, constraints, how to check? What did it get wrong?

Discuss: which gulf did each version help with, execution or evaluation?

ACTIVITY · 15 MINUTES · GROUPS

Try it: design for the second reader

Choose a site

A site you use every week: a department page, a course site, a library guide.

Write its llms.txt

One summary paragraph, then five links with one-line descriptions. What would an agent need first?

Add trust signals

For each link: what type of page, who wrote it, when it goes stale, and who checked it.

Discuss: what does a human reader lose or gain from the same signals?

REFERENCES · 1 OF 7

1. Weiser M. The computer for the 21st century. Sci Am. 1991;265(3):94–104. doi.org/10.1038/scientificamerican0991-94 (slide 3)
2. Ishii H, Ullmer B. Tangible bits: towards seamless interfaces between people, bits and atoms. In: Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems (CHI '97). New York: ACM; 1997. p. 234–241. doi.org/10.1145/258549.258715 (slide 3)
3. Subramonyam H, Pea R, Pondoc C, Agrawala M, Seifert C. Bridging the gulf of envisioning: cognitive challenges in prompt based interactions with LLMs. In: Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24). New York: ACM; 2024. p. 1–19. doi.org/10.1145/3613904.3642754 (slide 6)
4. Hutchins EL, Hollan JD, Norman DA. Direct manipulation interfaces. Hum Comput Interact. 1985;1(4):311–338. doi.org/10.1207/s15327051hci0104_2 (slide 6)
5. Riley J. Understanding metadata: what is metadata, and what is it for? A primer publication of the National Information Standards Organization. Baltimore: NISO; 2017. groups.niso.org/higherlogic/ws/public/download/17446/Understa... (slide 7)
6. Wilkinson MD, Dumontier M, Aalbersberg IJ, Appleton G, Axton M, Baak A, et al. The FAIR guiding principles for scientific data management and stewardship. Sci Data. 2016;3:160018. doi.org/10.1038/sdata.2016.18 (slide 7)
7. Hogan A, Blomqvist E, Cochez M, d'Amato C, de Melo G, Gutierrez C, et al. Knowledge graphs. ACM Comput Surv. 2021;54(4):71:1–71:37. doi.org/10.1145/3447772 (slide 7)
8. EMBL-EBI. Ontology Lookup Service (OLS4): NCBITaxon_41073 (Carabidae), hierarchical ancestors. Accessed 2026 Nov 13. ebi.ac.uk/ols4/ontologies/ncbitaxon/classes/http%253A%252F%25... (slide 7)
9. Edge D, Trinh H, Cheng N, Bradley J, Chao A, Mody A, et al. From local to global: a graph RAG approach to query-focused summarization. arXiv:2404.16130 [Preprint]. 2024. arxiv.org/abs/2404.16130 (slide 7)
10. Mitchell TM. The discipline of machine learning. Pittsburgh: Carnegie Mellon University; 2006. Report No.: CMU-ML-06-108. cs.cmu.edu/~tom/pubs/MachineLearning.pdf (slide 7)
11. Sutton RS, Barto AG. Reinforcement learning: an introduction. 2nd ed. Cambridge (MA): MIT Press; 2018. incompleteideas.net/book/RLbook2020.pdf (slide 7)
12. Ouyang L, Wu J, Jiang X, Almeida D, Wainwright CL, Mishkin P, et al. Training language models to follow instructions with human feedback. arXiv:2203.02155 [Preprint]. 2022. arxiv.org/abs/2203.02155 (slide 7)
13. OpenAI. OpenAI o1 system card. 2024 Dec 5. cdn.openai.com/o1-system-card-20241205.pdf (slide 7)
14. OpenAI. Addendum to o3 and o4-mini system card: Codex. 2025 May 16. cdn.openai.com/pdf/8df7697b-c1b2-4222-be00-1fd3298f351d/codex... (slide 7)

REFERENCES · 2 OF 7

15. Schluntz E, Zhang B. Building effective agents. Anthropic Engineering. 2024 Dec 19. anthropic.com/engineering/building-effective-agents (slide 7)

16. Swetnam TL. neon-beetle-seg: models and data dictionary (docs/models.md, docs/schema.md), v0.2.0. GitHub. 2026 Oct. github.com/tyson-swetnam/neon-beetle-seg/blob/main/docs/model... (slide 8)

17. Anthropic. Claude Code overview. Claude Code documentation. Accessed 2026 Nov 13. code.claude.com/docs/en/overview (slides 9, 38)

18. Google. Building the agentic future: developer highlights from I/O 2026. The Keyword. 2026 May 19. blog.google/innovation-and-ai/technology/developers-tools/goo... (slides 9, 38)

19. Anthropic. Let Claude use your computer in Cowork. Claude Help Center. Accessed 2026 Nov 13. support.claude.com/en/articles/14128542-let-claude-use-your-c... (slides 9, 38)

20. OpenAI. Computer use. ChatGPT and Codex documentation. Accessed 2026 Nov 13. learn.chatgpt.com/docs/computer-use (slides 9, 38)

21. Anthropic. Claude in Chrome [product page]. Accessed 2026 Nov 13. claude.com/chrome (slides 9, 38)

22. Google. The new era of browsing: putting Gemini to work in Chrome. The Keyword. 2026 Jan 28. blog.google/products-and-platforms/products/chrome/gemini-3-a... (slides 9, 38)

23. Perplexity. Comet [product page]. Accessed 2026 Nov 13. perplexity.ai/comet (slides 9, 38)

24. Anthropic. Use voice mode. Claude Help Center. Accessed 2026 Nov 13. support.claude.com/en/articles/11101966-use-voice-mode (slides 9, 38)

25. OpenAI. Voice. ChatGPT and Codex documentation. Accessed 2026 Nov 13. learn.chatgpt.com/docs/features/voice (slides 9, 38)

26. Google. Gemini gets more personal, proactive and powerful. The Keyword. 2025 May 20. blog.google/products-and-platforms/products/gemini/gemini-app... (slides 9, 38)

27. Anthropic. Continue local sessions from any device with Remote Control. Claude Code documentation. Accessed 2026 Nov 13. code.claude.com/docs/en/remote-control (slide 9)

28. OpenAI. Codex mobile. ChatGPT and Codex documentation. Accessed 2026 Nov 13. learn.chatgpt.com/docs/mobile (slides 9, 38)

29. Anthropic. Introducing Claude Duet. Claude blog. 2026 Oct 21. claude.com/blog/claude-duet (slides 9, 38)

30. OpenAI. dots: overview. ChatGPT and Codex documentation. Accessed 2026 Nov 13. learn.chatgpt.com/docs/dots?nav=overview (slides 9, 38)

31. Karpathy A. “There’s a new kind of coding I call ‘vibe coding’…” [post on X]. 2025 Feb 2. x.com/karpathy/status/1886192184808149383 (slide 11)

32. Collins Dictionary. Collins’ Word of the Year 2025: AI meets authenticity as society shifts. Collins Language Lovers blog. 2025 Nov 6. blog.collinsdictionary.com/language-lovers/collins-word-of-th… (slide 11)

33. METR. Measuring the impact of early-2025 AI on experienced open-source developer productivity. 2025 Jul 10. metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study (slide 12)

34. METR. We are changing our developer productivity experiment design. 2026 Feb 24. metr.org/blog/2026-02-24-uplift-update (slide 12)

35. Spracklen J, Wijewickrama R, Sakib AHMN, Maiti A, Viswanath B, Jadliwala M. We have a package for you! A comprehensive analysis of package hallucinations by code generating LLMs. In: 34th USENIX Security Symposium (USENIX Security ’25). 2025. arXiv:2406.10279. arxiv.org/abs/2406.10279 (slide 12)

36. Veracode. AI-generated code poses major security risks in nearly half of all development tasks, Veracode research reveals [press release on the 2025 GenAI Code Security Report]. 2025 Jul 30. veracode.com/press-release/ai-generated-code-poses-major-secu… (slide 12)

37. Radford A, Kim JW, Xu T, Brockman G, McLeavey C, Sutskever I. Robust speech recognition via large-scale weak supervision. In: Proceedings of the 40th International Conference on Machine Learning (ICML). PMLR 202; 2023. p. 28492–28518. arxiv.org/abs/2212.04356 (slide 13)

38. Anthropic. Voice dictation. Claude Code documentation. Accessed 2026 Nov 13. code.claude.com/docs/en/voice-dictation (slide 13)

39. Wispr. Wispr Flow features [vendor documentation]. Accessed 2026 Nov 13. wisprflow.ai/features (slide 13)

40. Wispr. Wispr Flow privacy [vendor documentation]. Accessed 2026 Nov 13. wisprflow.ai/privacy (slide 13)

41. superwhisper. Documentation: overview and coding agents [vendor documentation]. Accessed 2026 Nov 13. superwhisper.com/docs (slide 13)

42. Good Snooze (Bruin J). MacWhisper [vendor website]. Accessed 2026 Nov 13. macwhisper.com (slide 13)

43. Floridi L, Cowls J. A unified framework of five principles for AI in society. Harv Data Sci Rev. 2019;1(1). doi.org/10.1162/99608f92.8cd550d1 (slide 16)

44. White House Office of Science and Technology Policy. Blueprint for an AI Bill of Rights: making automated systems work for the American people. 2022 Oct. bidenwhitehouse.archives.gov/ostp/ai-bill-of-rights (slide 16)

45. International Energy Agency. AI is set to drive surging electricity demand from data centres while offering the potential to transform how the energy sector works [press release]. 2025 Apr 10. iea.org/news/ai-is-set-to-drive-surging-electricity-demand-fr... (slide 17)

46. Jevons WS. The coal question. London: Macmillan; 1865. Chapter VII. econlib.org/library/YPDBooks/Jevons/jvnCQ.html?chapter_num=9 (slide 17)

47. Bartz v. Anthropic PBC, No. 3:24-cv-05417 (N.D. Cal.). Order granting final approval of class action settlement. 2026 Jul 20. publishers.org/wp-content/uploads/2026/07/Order-Granting-Fina... (slide 17)

48. Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. Science. 2019;366(6464):447–453. doi.org/10.1126/science.aax2342 (slide 17)

49. METR, Redwood Research. Brief independent investigation of agents' behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident. 2026 Aug 26. metr.org/blog/2026-08-26-openai-hugging-face-incident-investi... (slide 17)

50. UNM Center for Advanced Research Computing. GPT 101: Ethics of artificial intelligence. 2026. CC BY 4.0. tyson-swetnam.github.io/intro-gpt/ethics (slide 17)

51. Bregman R. An inconvenient truth about AI. Substack. 2026 Jun 7. rutgerbregman.substack.com/p/an-inconvenient-truth-about-ai (slide 18)

52. Gates B. The turbulent AI era is here. The choices we make now are critical. Gates Notes. 2026 Aug 26. gatesnotes.com/a-turbulent-ai-era-and-critical-choices-to-make (slide 18)

53. Amodei D. We must pace the frontier. 2026 Sep. darioamodei.com/post/we-must-pace-the-frontier (slide 18)

54. Swetnam TL. How I learned to stop worrying and love the AI [UNM Computer Science Colloquium; slides and per-slide evidence]. 2026 Nov 4. tyson-swetnam.github.io/2026-11-04-cs-colloquium-/ (slides 19, 28, 30, 36, 39, 45, 46, 49)

55. Pope Leo XIV (@Pontifex). "The future of #ArtificialIntelligence is not pre-determined…" [post on X]. 2026 Oct 30. x.com/Pontifex/status/2116146977230459383 (slide 20)

56. Nelson A. "Red states and blue states are writing AI Bills of Rights" [post on Bluesky]. 2026 May 14. bsky.app/profile/alondra.bsky.social/post/3mltg2efhms2a (slide 20)

57. Howard J. /llms.txt: a proposal to provide information to help LLMs use websites. Answer.AI. 2024 Sep 3. answer.ai/posts/2024-09-03-llmstxt.html (slide 22)

58. Howard J. The /llms.txt file [proposal, version 2]. llmstxt.org. 2024 Sep 3; modified 2026 Aug 10. llmstxt.org (slide 22)

59. UNM Center for Advanced Research Computing. CARC documentation llms.txt. Accessed 2026 Nov 13. carc.unm.edu/docs/llms.txt (slides 22, 24)

60. MESA documentation. llms.txt [outline of the MESA MCP stack documentation]. idss-mesa/docs. Accessed 2026 Nov 13. idss-mesa.github.io/docs/llms.txt (slide 22)

61. UNM Center for Advanced Research Computing. DUST 2026: lessons on open science, research data management and AI ethics. llms.txt. Accessed 2026 Nov 13. unm-carc.github.io/dust-2026/llms.txt (slides 22, 27)

62. Google Cloud. Open Knowledge Format (OKF), version 0.2 [specification]. GoogleCloudPlatform/knowledge-catalog. github.com/GoogleCloudPlatform/knowledge-catalog/blob/main/ok... (slide 23)

63. UNM Center for Advanced Research Computing. CARC documentation: For AI agents. 2026 Sep 12 [accessed 2026 Nov 13]. carc.unm.edu/docs/about/ai-agents (slides 23, 24, 27)

64. MESA documentation. For AI agents. idss-mesa/docs. 2026 Oct 8 [accessed 2026 Nov 13]. idss-mesa.github.io/docs/about/ai-agents (slide 25)

65. Armbrust M, Ghodsi A, Xin R, Zaharia M. Lakehouse: a new generation of open platforms that unify data warehousing and advanced analytics. In: 11th Conference on Innovative Data Systems Research (CIDR '21). 2021. cidrdb.org/cidr2021/papers/cidr2021_paper17.pdf (slide 26)

66. Raasveldt M, Mühleisen H. DuckLake: SQL as a lakehouse format. DuckDB blog. 2025 May 27. duckdb.org/2025/05/27/ducklake.html (slide 26)

67. The DuckDB team. DuckLake 1.0: the lakehouse format built on SQL reaches production-readiness. 2026 Apr 13. ducklake.select/2026/04/13/ducklake-10 (slide 26)

68. MESA documentation. mesa-ducklake: the AVU metadata-history library that backs mesa-mcp. idss-mesa/docs. idss-mesa.github.io/docs/servers/mesa-ducklake (slide 26)

69. iRODS Consortium. iRODS 5.0.2 documentation: metadata (imeta). docs.irods.org/5.0.2/icommands/metadata (slide 26)

70. U.S. National Science Foundation. Award 2632685: Research Infrastructure: Category II: MESA (Multidisciplinary Environment for Scientific Advancement). 2026. nsf.gov/awardsearch/show-award/?AWD_ID=2632685 (slides 26, 30, 44)

REFERENCES · 6 OF 7

71. UNM Center for Advanced Research Computing. CARC documentation: Systems overview. Accessed 2026 Nov 13. carc.unm.edu/docs/systems/overview (slide 27)

72. UNM Center for Advanced Research Computing. CARC documentation: General FAQ. Accessed 2026 Nov 13. carc.unm.edu/docs/faq/general (slide 27)

73. UNM Center for Advanced Research Computing. CARC documentation: Getting started at CARC. Accessed 2026 Nov 13. carc.unm.edu/docs/getting-started/overview (slide 27)

74. UNM Center for Advanced Research Computing. CARC documentation: Getting help. Accessed 2026 Nov 13. carc.unm.edu/docs/support/help (slide 27)

75. UNM Center for Advanced Research Computing. CARC documentation: Workshops and slides. Accessed 2026 Nov 13. carc.unm.edu/docs/training/workshops (slide 27)

76. UNM Center for Advanced Research Computing. GPT 101: a workshop on generative AI and prompt engineering. 2026. CC BY 4.0. tyson-swetnam.github.io/intro-gpt (slides 27, 43)

77. UNM Center for Advanced Research Computing. AI research resources. Accessed 2026 Nov 13. carc.unm.edu/ai-research-resources (slide 27)

78. NVIDIA. DGX Spark hardware overview. NVIDIA documentation. Accessed 2026 Nov 13. docs.nvidia.com/dgx/dgx-spark/hardware.html (slide 28)

79. Shneiderman B. Direct manipulation: a step beyond programming languages. Computer. 1983;16(8):57–69. doi.org/10.1109/MC.1983.1654471 (slide 32)

80. Nielsen J. AI: first new UI paradigm in 60 years. Nielsen Norman Group. 2023 Jun 18. nngroup.com/articles/ai-paradigm (slide 32)

81. Shneiderman B. Direct manipulation for comprehensible, predictable and controllable user interfaces. 1997. Adapted from: Designing the user interface. 3rd ed. Reading (MA): Addison Wesley; 1997. cs.umd.edu/users/ben/papers/Shneiderman1997Direct2.pdf (slides 33, 40)

82. Anthropic. Introducing the Model Context Protocol. 2024 Nov 25. anthropic.com/news/model-context-protocol (slide 34)

83. Model Context Protocol. What is the Model Context Protocol (MCP)? Documentation. modelcontextprotocol.io/docs/getting-started/intro (slide 34)

84. Willison S. Not all AI-assisted programming is vibe coding (but vibe coding rocks). simonwillison.net. 2025 Mar 19. simonwillison.net/2025/Mar/19/vibe-coding (slide 35)

85. Karpathy A. [Post on X proposing “agentic engineering”]. 2026 Feb 4. x.com/karpathy/status/2019137879310836075 (slide 35)

86. Sarkar A, Drosos I. Vibe coding: programming through conversation with artificial intelligence. In: Proceedings of the 36th Annual Conference of the Psychology of Programming Interest Group (PPIG 2025). 2025. arXiv:2506.23253. arxiv.org/abs/2506.23253 (slide 35)

87. Center for Advanced Research Computing, University of New Mexico. AI Automation and Agents [online course]: AGENTS.md, the contract for agents editing the course site. 2026. CC BY 4.0. github.com/tyson-swetnam/AI-Automation-and-Agents/blob/main/A… (slide 36)

88. Ruan S, Wobbrock JO, Liou K, Ng A, Landay JA. Comparing speech and keyboard text entry for short messages in two languages on touchscreen phones. Proc ACM Interact Mob Wearable Ubiquitous Technol. 2017;1(4):159. doi.org/10.1145/3161187 (slide 37)

89. Nielsen J. The articulation barrier: prompt-driven AI UX hurts usability. Jakob Nielsen on UX. 2023 Jun 26. jakobnielsenphd.substack.com/p/prompt-driven-ai-ux-hurts-usab… (slide 37)

90. Nielsen J. Prompt augmentation: UX design patterns for better AI prompting. Jakob Nielsen on UX. 2025 Mar 27. jakobnielsenphd.substack.com/p/prompt-augmentation (slide 37)

91. Nowrin S, Ordóñez P, Vertanen K. Exploring motor-impaired programmers’ use of speech recognition. In: Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS ’22). New York: ACM; 2022. doi.org/10.1145/3517428.3550392 (slide 37)

92. Swetnam TL. neon-beetle-seg: segmenting and measuring NEON ground beetles [software and DuckLake lakehouse]. GitHub. github.com/tyson-swetnam/neon-beetle-seg (slide 39)

93. Amershi S, Weld D, Vorvoreanu M, Fourney A, Nushi B, Collisson P, et al. Guidelines for human-AI interaction. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI ’19). New York: ACM; 2019. p. 1–13. doi.org/10.1145/3290605.3300233 (slide 40)

94. Horvitz E. Principles of mixed-initiative user interfaces. In: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI ’99). New York: ACM; 1999. p. 159–166. doi.org/10.1145/302979.303030 (slide 40)

95. Barke S, James MB, Polikarpova N. Grounded Copilot: how programmers interact with code-generating models. Proc ACM Program Lang. 2023;7(OOPSLA1):85–111. doi.org/10.1145/3586030 (slide 40)

96. Swetnam TL. Guest lecture: Agentic AI loops: sequence, selection, iteration [slides]. AI Automation and Agents, University of New Mexico. 2026 Sep 22. CC BY 4.0. tyson-swetnam.github.io/AI-Automation-and-Agents/guest-lectur… (slides 41, 42)

97. Anthropic. Beyond permission prompts: making Claude Code more secure and autonomous. 2025 Oct 20. anthropic.com/engineering/claude-code-sandboxing (slide 43)

98. MESA: Multidisciplinary Environment for Scientific Advancement. NSF Award OAC-2632685. University of New Mexico. idss-mesa.github.io (slide 44)